

LENAWEE COUNTY AP, MI, USA

WMO: 725404

Lat: 41.868N

Lon: 84.079W

Elev: 243

StdP: 98.44

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.1	-22.1	0.5	-17.6	-18.9	0.7	-14.1	10.8	-2.4	9.7	-2.9	2.6	240	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.3	22.7	30.9	22.2	29.1	21.3	24.6	29.6	23.7	28.3	22.8	27.2	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.2	27.0	22.3	17.5	26.3	21.3	16.5	25.3	75.5	29.5	71.7	28.2	68.3	27.2	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.5	7.6	-21.9	34.2	3.9	1.8	-24.6	35.5	-26.9	36.6	-29.1	37.6	-31.9	39.0		
(5)			-22.1	25.8	3.6	0.9	-24.7	26.5	-26.8	27.0	-28.9	27.6	-31.5	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-4.2	-3.1	2.5	9.3	15.4	20.6	22.5	21.5	17.8	11.2	4.6	-1.3	(6)
(7)		DBStd	10.63	6.27	5.85	5.95	4.92	4.59	3.52	3.11	2.80	4.02	4.82	4.94	5.42	(7)
(8)		HDD10.0	1700	442	369	244	70	7	0	0	0	1	44	173	350	(8)
(9)		HDD18.3	3512	699	602	492	273	114	17	3	6	58	228	412	607	(9)
(10)		CDD10.0	1627	0	1	11	49	176	318	389	357	234	81	11	1	(10)
(11)		CDD18.3	398	0	0	1	2	24	85	134	106	41	6	0	0	(11)
(12)		CDH23.3	3533	0	0	6	34	257	741	1247	824	372	52	0	0	(12)
(13)		CDH26.7	1102	0	0	1	4	68	242	444	220	112	11	0	0	(13)
(14)	Wind	WSAvg	3.4	4.0	3.9	4.0	4.3	3.5	3.0	2.6	2.5	2.7	3.4	3.6	3.8	(14)
(15)	Precipitation	PrecAvg	845	47	43	60	79	84	91	80	87	84	59	70	63	(15)
(16)		PrecMax	1146	109	122	108	177	202	176	180	218	240	169	163	130	(16)
(17)		PrecMin	495	3	7	13	22	24	14	14	2	7	3	19	12	(17)
(18)		PrecStd	133	28	29	26	32	45	41	37	54	46	33	34	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.8	23.3	27.1	31.1	33.5	35.0	32.8	32.4	28.3	20.9	14.5	(19)
(20)			MCWB	11.5	11.4	17.6	18.0	20.5	22.6	23.6	23.3	21.7	21.1	14.7	12.7	(20)
(21)		2%	DB	9.8	10.9	17.9	23.8	28.5	31.4	32.5	30.6	29.8	24.4	17.2	11.5	(21)
(22)			MCWB	8.6	9.2	13.2	15.7	19.3	22.0	23.3	22.6	21.2	17.5	13.0	9.9	(22)
(23)		5%	DB	6.6	7.2	14.6	21.1	26.4	29.4	31.0	29.0	27.4	21.9	14.7	8.4	(23)
(24)			MCWB	4.9	5.2	10.9	13.8	18.8	20.9	22.5	21.8	19.7	16.6	11.4	6.9	(24)
(25)		10%	DB	3.7	4.1	11.7	18.0	23.8	27.5	29.1	27.6	25.2	19.3	12.4	6.0	(25)
(26)			MCWB	2.4	2.2	8.6	12.3	17.6	20.3	21.6	21.3	19.1	15.1	9.7	4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.5	17.8	19.5	22.6	24.5	25.8	25.6	24.3	21.6	16.4	13.0	(27)
(28)			MCDWB	12.9	14.0	23.4	24.4	28.3	30.6	31.7	30.5	29.6	26.9	19.2	14.1	(28)
(29)		2%	WB	8.7	8.8	14.2	17.2	21.0	23.3	24.8	24.4	22.8	19.4	13.8	10.1	(29)
(30)			MCDWB	9.5	10.6	17.1	21.3	26.5	28.9	30.1	28.2	27.1	22.2	16.3	11.3	(30)
(31)		5%	WB	5.2	5.1	11.7	15.0	19.9	22.5	23.8	23.3	21.4	17.4	11.9	7.1	(31)
(32)			MCDWB	6.6	6.8	13.9	19.9	24.5	27.7	28.8	26.9	25.3	20.3	13.9	8.3	(32)
(33)		10%	WB	2.5	2.4	8.7	13.0	18.7	21.4	22.9	22.3	20.2	15.6	9.8	4.5	(33)
(34)			MCDWB	3.6	4.0	11.4	17.6	22.7	25.9	27.2	25.9	23.8	18.9	12.3	5.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	9.3	11.3	11.3	11.0	11.4	10.9	11.8	10.5	8.6	6.7	(35)
(36)			MCDBR	9.4	11.0	13.9	15.5	14.4	13.4	12.9	12.3	14.4	13.9	12.3	9.1	(36)
(37)			MCWBR	8.4	9.1	9.9	9.1	7.3	6.0	5.3	5.7	6.8	7.8	8.9	8.1	(37)
(38)		5% WB	MCDBR	9.2	10.8	13.0	13.7	12.2	11.2	10.9	10.1	11.6	11.7	10.7	8.7	(38)
(39)			MCWBR	8.5	9.3	10.1	9.2	7.2	5.9	5.3	5.4	6.1	7.8	8.8	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.335	0.357	0.401	0.427	0.448	0.440	0.430	0.383	0.353	0.318	0.308	(40)	
(41)		taud	2.396	2.270	2.322	2.218	2.194	2.177	2.216	2.232	2.347	2.420	2.473	2.396	(41)	
(42)		Ebn at Noon	850	860	887	867	851	831	834	830	848	835	821	801	(42)	
(43)		Edh at Noon	80	106	114	136	143	146	139	133	110	91	74	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.53	2.44	3.53	4.61	5.38	6.03	6.08	5.29	4.30	2.82	1.79	1.25	(44)	
(45)		RadStd	0.16	0.33	0.33	0.41	0.48	0.40	0.32	0.22	0.37	0.25	0.24	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page